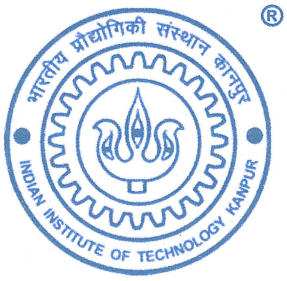
MTH 599

MSc Project II

Optimal Control and Viscosity Solutions of Hamilton-Jacobi-Bellman Equations



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Declaration

I hereby declare that the work presented in the report entitled *Optimal Control and Viscosity Solutions of Hamilton-Jacobi-Bellman Equations* is written by me in my own words and contains my own or borrowed ideas. At places where ideas and words are borrowed from other sources, proper references and acknowledgements, as applicable, have been provided. To the best of my knowledge this work does not emanate from or resemble work created by persons other than those mentioned and acknowledged herein.

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Abstract

This report studies the connection between optimal control problems and Hamilton-Jacobi-Bellman (HJB) equations. We consider the problem of minimizing a cost functional, consisting of a running cost and a terminal cost, over trajectories of a controlled ordinary differential equation, and introduce its value function. We prove that the value function is bounded and Lipschitz continuous, and that it satisfies the Dynamic Programming Principle. From this principle we derive the HJB equation, under the assumption that the value function is C^1, and show how a smooth solution yields an optimal feedback control.

Since the value function is in general not C^1, this classical derivation is insufficient. We discuss why almost everywhere solutions are not unique, and introduce the notion of viscosity solution. We verify that this notion is consistent with classical solutions, treat terminal-value problems by time reversal, and prove uniqueness of viscosity solutions by the doubling-of-variables technique. Combining these results, we show that the value function is the unique viscosity solution of the terminal-value problem for the HJB equation. Finally, we discuss which part of the construction of optimal feedback controls survives when the value function is only Lipschitz continuous.

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Chapter 1

Optimal Control Theory: A crash course

1.1 Important Terminology

Definition 1.1 ([Eva10]). Let $A \subset \mathbb{R}^m$ be a nonempty compact set, called *the control set*, and let $f : \mathbb{R}^n \times A \to \mathbb{R}^n$ be a bounded, continuous function that is uniformly Lipschitz continuous in its first variable. That is, there exist constants $M, L > 0$ such that, for all $x, y \in \mathbb{R}^n$ and $a \in A$,

$$|f(x,a)| \leq M, \qquad |f(x,a) - f(y,a)| \leq L|x-y|. \tag{1.1}$$

For a fixed time horizon $[t,T]$ an *admissible control* is defined as a Lebesgue-measurable function $\alpha : [t,T] \to A$. We denote the set of admissible controls by $\mathcal{A}$.

Given an initial state $x \in \mathbb{R}^n$ at time t and a control $\alpha \in \mathcal{A}$, consider the dynamical system

$$\begin{cases} \dot{\mathbf{x}}(s) = f(\mathbf{x}(s), \alpha(s)) & s \in [t,T] \\ \mathbf{x}(t) = x. \end{cases} \tag{1.2}$$

Here, $\mathbf{x}(s) \in \mathbb{R}^n$ denotes the *state* at time s, while $\alpha(s)$ denotes the control applied at that time. The trajectory $\mathbf{x}(\cdot)$ is the *response* of the system to the control $\alpha(\cdot)$. Hence (1.2) is called a *controlled dynamical system.*

One natural question that arises is the existence and uniqueness of $\mathbf{x}(\cdot)$, which is guaranteed by the following theorem from ODE theory, which we state without proof:

Theorem 1.2 (CARATHEODORY EXISTENCE AND UNIQUENESS [Rou05]). *Let $I = [t,T]$ and let $g : I \times \mathbb{R}^n \to \mathbb{R}^n$ be a Caratheodory mapping, that is, for every fixed $y \in \mathbb{R}^n$, $s \mapsto g(s,y)$ is measurable and, for almost every $s \in I$, $y \mapsto g(s,y)$ is continuous. Suppose further that g satisfies the growth condition $|g(s,y)| \leq \gamma(s) + C|y|$ for some $\gamma \in L^1(I)$ and $C \geq 0$. Then the initial-value problem*

$$\begin{cases} \dot{y}(s) = g(s, y(s)), & s \in I, \\ y(t) = x \end{cases}$$

admits an absolutely continuous solution $y : I \to \mathbb{R}^n$. If, in addition, there exists $\beta \in L^1(I)$ such that

$$|g(s,y_1) - g(s,y_2)| \leq \beta(s)|y_1 - y_2|$$

for almost every $s \in I$ and all $y_1, y_2 \in \mathbb{R}^n$, then the solution is unique.

We apply the theorem to the controlled dynamical system (1.2). For a fixed admissible control $\alpha \in \mathcal{A}$, define

$$g(s, y) := f(y, \alpha(s)).$$

Since α is measurable and f is continuous, g is measurable in s for every fixed y and continuous in y for every fixed s. Hence g is a Carathéodory mapping.

Moreover, by the boundedness of f,

$$|g(s, y)| = |f(y, \alpha(s))| \leq M.$$

Thus, the growth condition holds with $\gamma(s) = M$ and $C = 0$, where $\gamma \in L^1([t, T])$.

Finally, the uniform Lipschitz continuity of f in its first variable gives

$$\begin{aligned} |g(s, y_1) - g(s, y_2)| &= |f(y_1, \alpha(s)) - f(y_2, \alpha(s))| \\ &\leq L|y_1 - y_2|. \end{aligned}$$

Taking $\beta(s) = L$, we have $\beta \in L^1([t, T])$.

All the hypotheses of the theorem are satisfied. Consequently, for every initial state $x \in \mathbb{R}^n$ and every admissible control $\alpha \in \mathcal{A}$, the controlled dynamical system (1.2) admits a unique absolutely continuous solution $\mathbf{x}(\cdot) : [t, T] \to \mathbb{R}^n$.

Our goal is to find a control $\alpha^*(\cdot)$ which optimally steers the system. However, in order to define what "optimal" means, we must first introduce a *cost criterion*.

Definition 1.3 (COST FUNCTIONAL [Eva10])**.** Given $x \in \mathbb{R}^n$ and $0 \leq t \leq T$, for each admissible control $\alpha(\cdot) \in \mathcal{A}$, we define the corresponding *cost functional* as

$$C_{x,t}[\alpha(\cdot)] := \int_t^T r(\mathbf{x}(s), \alpha(s))\, ds + g(\mathbf{x}(T)), \tag{1.3}$$

where $\mathbf{x}(\cdot)$ solves the ODE (1.2) and $r : \mathbb{R}^n \times A \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}$ are given functions, called the *running cost per unit time* and the *terminal cost*, respectively.

Henceforth, the following will be our standing assumption:

$$\begin{cases} |r(x, a)| \leq C & |g(x)| \leq C \\ |r(x, a) - r(y, a)| \leq C|x - y| & |g(x) - g(y)| \leq C|x - y|. \end{cases} \tag{1.4}$$

We also assume that r is continuous on $\mathbb{R}^n \times A$, so that the minima over the compact set A appearing below are attained.

Now we can state our goal clearly: finding an optimal control for the system (1.2) means finding a control $\alpha^*(\cdot)$ that minimizes the cost functional starting at (x, t). Let us define the minimum cost incurred when starting from (x, t).

Definition 1.4 (VALUE FUNCTION [Eva10])**.** Given the system (1.2), starting at (x, t) ($x \in \mathbb{R}^n$, $0 \leq t \leq T$) we define the *value function* as

$$u(x, t) = \inf_{\alpha(\cdot) \in \mathcal{A}} C_{x,t}[\alpha(\cdot)]. \tag{1.5}$$

Our plan is to find a necessary condition on $u(x, t)$, which we later find to be the *Hamilton-Jacobi-Bellman* (HJB) equation, and then to use the value function and the HJB equation to design an optimal control.

But first let us deduce some properties of the value function u. We claim that u is bounded and Lipschitz continuous in both variables. In order to prove this claim, let us recall an inequality from standard ODE theory, which we state without proof:

Lemma 1.5 (GRONWALL INEQUALITY (Integral Form) [Eva10]). *Let $\xi(s)$ be a non-negative, summable function on $[t,T]$ which satisfies the integral inequality*

$$\xi(s) \le C_1 \int_t^s \xi(r)\,dr + C_2$$

for almost every $s \in [t,T]$, for constants $C_1, C_2 \ge 0$. Then

$$\xi(s) \le C_2\Big(1 + C_1(s-t)e^{C_1(s-t)}\Big)$$

for almost every $s \in [t,T]$.

We use this lemma to find estimates of u:

Theorem 1.6 (ESTIMATES FOR THE VALUE FUNCTION [Eva10]). *There exists a constant C such that*

$$|u(x,t)| \le C,$$
$$|u(x,t) - u(\hat{x},\hat{t})| \le C(|x-\hat{x}| + |t-\hat{t}|)$$

for all $x, \hat{x} \in \mathbb{R}^n$, and $t, \hat{t} \in [0,T]$.

Proof. Throughout this proof, K denotes a constant which depends only on T, M, L and the constant C of (1.4), and which may change from line to line.

Step 1: Boundedness. By (1.4), for every $\alpha \in \mathcal{A}$,

$$|C_{x,t}[\alpha]| \le \int_t^T |r(\mathbf{x}(s),\alpha(s))|\,ds + |g(\mathbf{x}(T))| \le C(T-t) + C \le C(T+1),$$

so $|u(x,t)| \le C(T+1)$ for all (x,t).

Step 2: Lipschitz continuity in x. Fix $x, \hat{x} \in \mathbb{R}^n$ and $0 \le t < T$ (for $t = T$ we have $u(\cdot,T) = g$, which is Lipschitz by (1.4)). Let $\varepsilon > 0$ and choose $\hat{\alpha} \in \mathcal{A}$ such that

$$u(\hat{x},t) + \varepsilon \ge \int_t^T r(\hat{\mathbf{x}}(s),\hat{\alpha}(s))\,ds + g(\hat{\mathbf{x}}(T)), \tag{1.6}$$

where $\hat{\mathbf{x}}(\cdot)$ solves

$$\begin{cases} \dot{\hat{\mathbf{x}}}(s) = f(\hat{\mathbf{x}}(s),\hat{\alpha}(s)) \quad (t < s < T), \\ \hat{\mathbf{x}}(t) = \hat{x}. \end{cases} \tag{1.7}$$

Since $\hat{\alpha} \in \mathcal{A}$ is also admissible for the initial point x, we get

$$\begin{aligned} u(x,t) - u(\hat{x},t) \le &\int_t^T r(\mathbf{x}(s),\hat{\alpha}(s))\,ds + g(\mathbf{x}(T)) \\ &- \int_t^T r(\hat{\mathbf{x}}(s),\hat{\alpha}(s))\,ds - g(\hat{\mathbf{x}}(T)) + \varepsilon, \end{aligned} \tag{1.8}$$

where $\mathbf{x}(\cdot)$ solves

$$\begin{cases} \dot{\mathbf{x}}(s) = f(\mathbf{x}(s),\hat{\alpha}(s)) \quad (t < s < T), \\ \mathbf{x}(t) = x. \end{cases} \tag{1.9}$$

Subtracting (1.7) from (1.9) and using the Lipschitz continuity of f,

$$|\mathbf{x}(s) - \hat{\mathbf{x}}(s)| \le |x - \hat{x}| + L\int_t^s |\mathbf{x}(\rho) - \hat{\mathbf{x}}(\rho)|\,d\rho.$$

Gronwall's inequality (Lemma 1.5) with $C_1 = L$, $C_2 = |x - \hat{x}|$ gives

$$|\mathbf{x}(s) - \hat{\mathbf{x}}(s)| \le |x - \hat{x}|\Big(1 + L(s-t)e^{L(s-t)}\Big) \le K|x - \hat{x}| \qquad (t \le s \le T).$$

Hence (1.4) and (1.8) yield $u(x,t) - u(\hat{x},t) \le K|x - \hat{x}| + \varepsilon$. Letting $\varepsilon \to 0$ and exchanging the roles of x and $\hat{x}$,

$$|u(x,t) - u(\hat{x},t)| \le K|x - \hat{x}| \qquad (x, \hat{x} \in \mathbb{R}^n,\ 0 \le t \le T).$$

Step 3: Lipschitz continuity in t. Let $x \in \mathbb{R}^n$ and $0 \le t < \hat{t} \le T$.

(a) The bound for $u(x,\hat{t}) - u(x,t)$. Take $\varepsilon > 0$ and choose $\alpha \in \mathcal{A}$ such that

$$u(x,t) + \varepsilon \ge \int_t^T r(\mathbf{x}(s), \alpha(s))\, ds + g(\mathbf{x}(T)),$$

where $\mathbf{x}(\cdot)$ solves (1.2) with $\mathbf{x}(t) = x$. Define the shifted control

$$\hat{\alpha}(s) := \alpha(s + t - \hat{t}) \qquad (\hat{t} \le s \le T),$$

and let $\hat{\mathbf{x}}(\cdot)$ solve

$$\begin{cases} \dot{\hat{\mathbf{x}}}(s) = f(\hat{\mathbf{x}}(s), \hat{\alpha}(s)) & (\hat{t} < s < T), \\ \hat{\mathbf{x}}(\hat{t}) = x. \end{cases}$$

Since the equation does not depend explicitly on time, $\hat{\mathbf{x}}(s) = \mathbf{x}(s + t - \hat{t})$. Hence

$$\begin{aligned} u(x,\hat{t}) - u(x,t) &\le \int_{\hat{t}}^T r(\hat{\mathbf{x}}(s), \hat{\alpha}(s))\, ds + g(\hat{\mathbf{x}}(T)) - \int_t^T r(\mathbf{x}(s), \alpha(s))\, ds - g(\mathbf{x}(T)) + \varepsilon \\ &= -\int_{T+t-\hat{t}}^T r(\mathbf{x}(s), \alpha(s))\, ds + g(\mathbf{x}(T + t - \hat{t})) - g(\mathbf{x}(T)) + \varepsilon \\ &\le K|t - \hat{t}| + \varepsilon, \end{aligned}$$

because $|r| \le C$ and, since $|\dot{\mathbf{x}}| \le M$, $|g(\mathbf{x}(T + t - \hat{t})) - g(\mathbf{x}(T))| \le CM|t - \hat{t}|$.

(b) The bound for $u(x,t) - u(x,\hat{t})$. Choose $\hat{\alpha} \in \mathcal{A}$ (on $[\hat{t}, T]$) such that

$$u(x,\hat{t}) + \varepsilon \ge \int_{\hat{t}}^T r(\hat{\mathbf{x}}(s), \hat{\alpha}(s))\, ds + g(\hat{\mathbf{x}}(T)),$$

where $\hat{\mathbf{x}}(\cdot)$ solves $\dot{\hat{\mathbf{x}}} = f(\hat{\mathbf{x}}, \hat{\alpha})$ for $\hat{t} < s < T$, $\hat{\mathbf{x}}(\hat{t}) = x$. Fix any $a_0 \in A$ and define

$$\alpha(s) := \begin{cases} \hat{\alpha}(s + \hat{t} - t) & \text{if } t \le s \le T + t - \hat{t}, \\ a_0 & \text{if } T + t - \hat{t} < s \le T, \end{cases}$$

and let $\mathbf{x}(\cdot)$ solve (1.2) with $\mathbf{x}(t) = x$. Then $\mathbf{x}(s) = \hat{\mathbf{x}}(s + \hat{t} - t)$ for $t \le s \le T + t - \hat{t}$, and so

$$\begin{aligned} u(x,t) - u(x,\hat{t}) &\le \int_t^T r(\mathbf{x}(s), \alpha(s))\, ds + g(\mathbf{x}(T)) - \int_{\hat{t}}^T r(\hat{\mathbf{x}}(s), \hat{\alpha}(s))\, ds - g(\hat{\mathbf{x}}(T)) + \varepsilon \\ &= \int_{T+t-\hat{t}}^T r(\mathbf{x}(s), \alpha(s))\, ds + g(\mathbf{x}(T)) - g(\mathbf{x}(T + t - \hat{t})) + \varepsilon \\ &\le K|t - \hat{t}| + \varepsilon, \end{aligned}$$

where we used $\hat{\mathbf{x}}(T) = \mathbf{x}(T + t - \hat{t})$.

Letting $\varepsilon \to 0$ in (a) and (b),

$$|u(x,t) - u(x,\hat{t})| \le K|t - \hat{t}| \qquad (0 \le t \le \hat{t} \le T,\ x \in \mathbb{R}^n).$$

Step 4: Conclusion. By the triangle inequality, Step 2 (at time $\hat{t}$) and Step 3 (at the point x),

$$|u(x,t) - u(\hat{x},\hat{t})| \le |u(x,t) - u(x,\hat{t})| + |u(x,\hat{t}) - u(\hat{x},\hat{t})| \le K\Big(|t - \hat{t}| + |x - \hat{x}|\Big). \quad \square$$

1.2 Dynamic Programming Principle

The central idea underlying the Dynamic Programming Principle is that an optimal control must remain optimal on every subsequent subinterval along its trajectory. More precisely, if a control is optimal over a given time horizon, then, after any intermediate time, its restriction to the remaining interval must be optimal for the problem starting from the state reached at that time. Otherwise, one could replace the remaining portion of the control by a better admissible control, thereby improving the original one and contradicting its optimality.

Consequently, the optimal cost from a given initial state can be decomposed into two contributions: the cost accumulated up to an arbitrary intermediate time and the optimal cost-to-go from the state reached at that time. This observation constitutes the Dynamic Programming Principle.

Now we formalise the above statements mathematically and prove them for our value function:

Theorem 1.7 (VALUE FUNCTION SATISFIES DPP [Eva])**.** *For each $h > 0$ small enough that $t + h \le T$, we have*

$$u(x,t) = \inf_{\alpha \in \mathcal{A}} \left\{ \int_t^{t+h} r(\mathbf{x}(s), \alpha(s))\, ds + u(\mathbf{x}(t+h), t+h) \right\}, \tag{1.10}$$

where $\mathbf{x}(\cdot) = \mathbf{x}^{\alpha}(\cdot)$ solves the ODE (1.2) for the control $\alpha(\cdot)$, and $u(\cdot,\cdot)$ is the value function.

Proof. **Step 1: "$\le$".** Choose any control $\alpha_1 \in \mathcal{A}$ (only its values on $[t, t+h]$ matter below) and solve the ODE

$$\begin{cases} \dot{\mathbf{x}}_1(s) = f(\mathbf{x}_1(s), \alpha_1(s)) & (t < s < t+h), \\ \mathbf{x}_1(t) = x. \end{cases}$$

Fix $\varepsilon > 0$ and choose $\alpha_2 \in \mathcal{A}$ (on $[t+h, T]$) such that

$$u(\mathbf{x}_1(t+h), t+h) + \varepsilon \ge \int_{t+h}^{T} r(\mathbf{x}_2(s), \alpha_2(s))\, ds + g(\mathbf{x}_2(T)), \tag{1.11}$$

where

$$\begin{cases} \dot{\mathbf{x}}_2(s) = f(\mathbf{x}_2(s), \alpha_2(s)) & (t+h < s < T), \\ \mathbf{x}_2(t+h) = \mathbf{x}_1(t+h). \end{cases}$$

Now define the control

$$\alpha_3(s) := \begin{cases} \alpha_1(s) & \text{if } t \le s < t+h, \\ \alpha_2(s) & \text{if } t+h \le s \le T, \end{cases}$$

and let $\mathbf{x}_3(\cdot)$ solve

$$\begin{cases} \dot{\mathbf{x}}_3(s) = f(\mathbf{x}_3(s), \alpha_3(s)) & (t < s < T), \\ \mathbf{x}_3(t) = x. \end{cases}$$

By uniqueness of solutions of the ODE (1.2),

$$\mathbf{x}_3(s) = \begin{cases} \mathbf{x}_1(s) & \text{if } t \le s \le t+h, \\ \mathbf{x}_2(s) & \text{if } t+h \le s \le T. \end{cases}$$

Thus the definition (1.5) of u implies

$$\begin{aligned} u(x,t) \le C_{x,t}[\alpha_3] &= \int_t^T r(\mathbf{x}_3(s),\alpha_3(s))\,ds + g(\mathbf{x}_3(T)) \\ &= \int_t^{t+h} r(\mathbf{x}_1(s),\alpha_1(s))\,ds + \int_{t+h}^T r(\mathbf{x}_2(s),\alpha_2(s))\,ds + g(\mathbf{x}_2(T)) \\ &\le \int_t^{t+h} r(\mathbf{x}_1(s),\alpha_1(s))\,ds + u(\mathbf{x}_1(t+h),t+h) + \varepsilon, \end{aligned}$$

the last inequality resulting from (1.11). As $\alpha_1 \in \mathcal{A}$ was arbitrary, we conclude

$$u(x,t) \le \inf_{\alpha\in\mathcal{A}} \left\{ \int_t^{t+h} r(\mathbf{x}(s),\alpha(s))\,ds + u(\mathbf{x}(t+h),t+h) \right\} + \varepsilon, \tag{1.12}$$

where $\mathbf{x}(\cdot) = \mathbf{x}^\alpha(\cdot)$ solves (1.2).

Step 2: "$\ge$". Fix again $\varepsilon > 0$ and select $\alpha_4 \in \mathcal{A}$ such that

$$u(x,t) + \varepsilon \ge \int_t^T r(\mathbf{x}_4(s),\alpha_4(s))\,ds + g(\mathbf{x}_4(T)), \tag{1.13}$$

where

$$\begin{cases} \dot{\mathbf{x}}_4(s) = f(\mathbf{x}_4(s),\alpha_4(s)) \quad (t < s < T), \\ \mathbf{x}_4(t) = x. \end{cases}$$

The restriction of α_4 to $[t+h,T]$ is admissible for the initial point $\mathbf{x}_4(t+h)$ at time $t+h$, so by (1.5)

$$u(\mathbf{x}_4(t+h),t+h) \le \int_{t+h}^T r(\mathbf{x}_4(s),\alpha_4(s))\,ds + g(\mathbf{x}_4(T)). \tag{1.14}$$

Therefore, by (1.13) and (1.14),

$$\begin{aligned} u(x,t) + \varepsilon &\ge \int_t^{t+h} r(\mathbf{x}_4(s),\alpha_4(s))\,ds + u(\mathbf{x}_4(t+h),t+h) \\ &\ge \inf_{\alpha\in\mathcal{A}} \left\{ \int_t^{t+h} r(\mathbf{x}(s),\alpha(s))\,ds + u(\mathbf{x}(t+h),t+h) \right\}. \end{aligned}$$

This inequality and (1.12), with $\varepsilon \to 0$, complete the proof of (1.10). □

1.3 Derivation of the *Hamilton- Jacobi-Bellman* Equation

From this section onwards we use the following notation: for a function $u : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}$, we write u_t for $\frac{\partial u}{\partial t}$ and $Du = (\frac{\partial u}{\partial x_1}, \dots, \frac{\partial u}{\partial x_n})$ for the gradient in the space variable; $a \cdot b$ denotes the Euclidean inner product.

In any type of optimization problem, we are interested in finding a necessary condition. In calculus, *Fermat's theorem on stationary points* states that for a differentiable function, if an extremum is attained in the interior, then the derivative of the function at that point must be zero. Similarly, in the calculus of variations, an optimal curve must satisfy the *Euler-Lagrange equation.*

In the same spirit, we assert that for an optimal control problem the *value function* must satisfy the *Hamilton-Jacobi-Bellman equation.*

Theorem 1.8 (VALUE FUNCTION SOLVES HJB [Eva]). *Assume that the value function u is a C^1 function of the variables (x,t). Then u solves the terminal value problem*

$$\begin{cases} u_t + \min_{a\in A}\{f(x,a)\cdot Du(x,t) + r(x,a)\} = 0 \quad (x\in\mathbb{R}^n, 0\le t<T), \\ u(x,T) = g(x). \end{cases} \tag{1.15}$$

Proof. The terminal condition holds because the integral in (1.3) is over an empty interval when $t=T$, so $C_{x,T}[\alpha] = g(x)$ for every α. Now fix $x\in\mathbb{R}^n$, $0\le t<T$ and $0<h<T-t$. Put

$$\Theta(y,s) := u_t(y,s) + \min_{a\in A}\{f(y,a)\cdot Du(y,s) + r(y,a)\}.$$

Since $u\in C^1$, f and r are continuous and A is compact, Θ is well defined and continuous. We show $\Theta(x,t)=0$.

Step 1: $\Theta(x,t)\ge 0$. Pick any $a\in A$ and use the constant control $\alpha\equiv a$ on $[t,T]$. Let $\mathbf{x}(\cdot)$ be its response,

$$\dot{\mathbf{x}}(s) = f(\mathbf{x}(s),a), \qquad \mathbf{x}(t) = x.$$

This control is admissible in (1.10), so Theorem 1.7 gives

$$u(x,t) \le \int_t^{t+h} r(\mathbf{x}(s),a)\,ds + u(\mathbf{x}(t+h),t+h). \tag{1.16}$$

Step 2: We rearrange (1.16) and divide by $h>0$:

$$\frac{u(\mathbf{x}(t+h),t+h) - u(x,t)}{h} + \frac{1}{h}\int_t^{t+h} r(\mathbf{x}(s),a)\,ds \ge 0.$$

Since f is continuous and the control is constant, $\mathbf{x}(\cdot)$ is C^1 with $\dot{\mathbf{x}}(t) = f(x,a)$. As $u\in C^1$ and $s\mapsto r(\mathbf{x}(s),a)$ is continuous, letting $h\to 0$ gives

$$u_t(x,t) + Du(x,t)\cdot\dot{\mathbf{x}}(t) + r(x,a) \ge 0,$$

that is, $u_t(x,t) + f(x,a)\cdot Du(x,t) + r(x,a)\ge 0$. This holds for every $a\in A$, hence

$$\min_{a\in A}\{u_t(x,t) + f(x,a)\cdot Du(x,t) + r(x,a)\} \ge 0, \qquad \text{i.e. } \Theta(x,t)\ge 0. \tag{1.17}$$

Step 3: The minimum in (1.17) equals zero. By the definition of the infimum, there is a control $\alpha_h\in\mathcal{A}$ with

$$C_{x,t}[\alpha_h] \le u(x,t) + h^2.$$

Let $\mathbf{x}_h(\cdot)$ be its response. As in Step 2 of the proof of Theorem 1.7, $u(\mathbf{x}_h(t+h),t+h) \le \int_{t+h}^T r(\mathbf{x}_h,\alpha_h)\,ds + g(\mathbf{x}_h(T))$, and therefore

$$u(x,t) + h^2 \ge \int_t^{t+h} r(\mathbf{x}_h(s),\alpha_h(s))\,ds + u(\mathbf{x}_h(t+h),t+h).$$

Rearranging and dividing by h,

$$h \ge \frac{u(\mathbf{x}_h(t+h),t+h) - u(x,t)}{h} + \frac{1}{h}\int_t^{t+h} r(\mathbf{x}_h(s),\alpha_h(s))\,ds. \tag{1.18}$$

The response $\mathbf{x}_h$ is Lipschitz ($|\dot{\mathbf{x}}_h| \le M$) and $u \in C^1$, so $s \mapsto u(\mathbf{x}_h(s), s)$ is absolutely continuous with derivative $u_t + Du \cdot f(\mathbf{x}_h, \alpha_h)$ almost everywhere. Hence the right-hand side of (1.18) equals

$$\frac{1}{h}\int_t^{t+h} \Big[u_t(\mathbf{x}_h(s), s) + f(\mathbf{x}_h(s), \alpha_h(s)) \cdot Du(\mathbf{x}_h(s), s) + r(\mathbf{x}_h(s), \alpha_h(s))\Big] ds$$
$$\ge \frac{1}{h}\int_t^{t+h} \Theta(\mathbf{x}_h(s), s)\, ds.$$

Since $|\mathbf{x}_h(s) - x| \le Mh$ for $t \le s \le t+h$ and Θ is continuous, $\Theta(\mathbf{x}_h(s), s) \to \Theta(x, t)$ uniformly in $s \in [t, t+h]$ as $h \to 0$. Letting $h \to 0$ in (1.18) gives $0 \ge \Theta(x, t)$. Together with (1.17), $\Theta(x, t) = 0$, which is (1.15). □

1.4 Design of Optimal Controls

Now that we have a necessary condition, we exploit the Hamilton-Jacobi-Bellman equation and the value function to design an optimal control whose cost is the value function [Eva]:

Step 1: Solve the Hamilton–Jacobi–Bellman equation, and thereby compute the value function u.

Step 2: Use the value function u and the Hamilton–Jacobi–Bellman PDE to design an optimal feedback control $\alpha^*(\cdot)$, as follows. Fix a point $x \in \mathbb{R}^n$ and a time $0 \le t \le T$. The map $\{a \mapsto f(x, a) \cdot Du(x, t) + r(x, a)\}$ is continuous and A is a compact subset of $\mathbb{R}^m$, so the map attains its minimum over A. Let $a_{x,t}$ be a point where the minimum is attained, and define $\alpha(x, t) := a_{x,t}$, a parameter value at which the minimum in (1.15) is attained. For this $\alpha(x, t)$, (1.15) becomes

$$u_t(x, t) + f(x, \alpha(x, t)) \cdot Du(x, t) + r(x, \alpha(x, t)) = 0. \tag{1.19}$$

Next we solve the following ODE, assuming $\alpha(\cdot, \cdot)$ is sufficiently regular to let us do so:

$$\begin{cases} \dot{\mathbf{x}}^*(s) = f(\mathbf{x}^*(s), \alpha(\mathbf{x}^*(s), s)) \quad (t \le s \le T), \\ \mathbf{x}^*(t) = x. \end{cases} \tag{1.20}$$

Finally, define the *feedback control*

$$\alpha^*(s) := \alpha(\mathbf{x}^*(s), s). \tag{1.21}$$

We demonstrate next that this construction does indeed provide us with an optimal control.

Theorem 1.9 (VERIFICATION OF OPTIMALITY [Eva])**.** *The control α^* defined by the construction (1.21) is optimal.*

Proof. From (1.3) we have

$$C_{x,t}[\alpha^*(\cdot)] = \int_t^T r(\mathbf{x}^*(s), \alpha^*(s))\, ds + g(\mathbf{x}^*(T)). \tag{1.22}$$

Given $\alpha^*(s) := \alpha(\mathbf{x}^*(s), s)$, and since $\alpha(\mathbf{x}^*(s), s)$ satisfies (1.19), we get

$$u_t(\mathbf{x}^*(s), s) + f(\mathbf{x}^*(s), \alpha^*(s)) \cdot Du(\mathbf{x}^*(s), s) + r(\mathbf{x}^*(s), \alpha^*(s)) = 0.$$

Substituting this expression for $r(\mathbf{x}^*(s), \alpha^*(s))$ into (1.22) and using (1.20), we get

$$\begin{aligned}
C_{x,t}[\alpha^*(\cdot)] &= -\int_t^T \Big(u_t(\mathbf{x}^*(s), s) + Du(\mathbf{x}^*(s), s) \cdot \dot{\mathbf{x}}^*(s) \Big)\, ds \\
&\quad + g(\mathbf{x}^*(T)) \\
&= -\int_t^T \frac{d}{ds} u(\mathbf{x}^*(s), s)\, ds + g(\mathbf{x}^*(T)) && \text{(chain rule)} \\
&= -u(\mathbf{x}^*(T), T) + u(\mathbf{x}^*(t), t) + g(\mathbf{x}^*(T)) && \text{(integrating)} \\
&= -g(\mathbf{x}^*(T)) + u(x, t) + g(\mathbf{x}^*(T)) && \text{(terminal cond.)} \\
&= u(x, t) = \inf_{\alpha(\cdot) \in \mathcal{A}} C_{x,t}[\alpha(\cdot)].
\end{aligned}$$

□

Chapter 2

Viscosity solutions

Both Theorem 1.8 and Theorem 1.9 rely on the strong assumption that $u \in C^1$, but in practice we need not encounter $u \in C^1$.

Example 2.1. Fix $T > t \geq 0$ and consider the problem in one space dimension

$$\min_{\alpha(\cdot)} \int_t^T |\alpha(s)|\, ds \qquad \text{subject to} \qquad \dot{\mathbf{x}}(s) = \alpha(s), \quad \mathbf{x}(t) = x, \quad \mathbf{x}(T) = 0,$$

where $\alpha : [t, T] \to \mathbb{R}$ is a measurable control with $\alpha \in L^1([t, T])$. Here $f(x, a) = a$ and $r(x, a) = |a|$, and the terminal condition is imposed as a constraint. (This example lies outside the standing assumptions of Chapter 1: the control set $\mathbb{R}$ is not compact, r is unbounded, and the terminal condition is a hard constraint rather than a Lipschitz terminal cost. It still shows that the value function need not be C^1.)

The value function is

$$u(x, t) = \inf_{\alpha} \Big\{ \int_t^T |\alpha(s)|\, ds : \ \mathbf{x}(t) = x, \ \mathbf{x}(T) = 0 \Big\}.$$

Since $\dot{\mathbf{x}} = \alpha$, the terminal constraint reads $\int_t^T \alpha(s)\, ds = -x$. By the triangle inequality,

$$\int_t^T |\alpha(s)|\, ds \ \geq \ \Big| \int_t^T \alpha(s)\, ds \Big| = |x|.$$

This lower bound is attained by the constant control $\alpha^*(s) = -\frac{x}{T-t}$, which satisfies the constraint and has cost

$$\int_t^T |\alpha^*(s)|\, ds = \int_t^T \frac{|x|}{T-t}\, ds = |x|.$$

Consequently

$$u(x, t) = |x| \qquad (x \in \mathbb{R}, \ 0 \leq t < T).$$

This function is Lipschitz, but it is not differentiable at $x = 0$, so $u \notin C^1$ and Theorem 1.8 cannot be applied to it.

The value function exists regardless of whether it is C^1 or not, so we may ask whether the value function solves the Hamilton-Jacobi-Bellman equation in some weaker sense. In order to answer this, we must first define a weaker notion of solution of the Hamilton-Jacobi-Bellman equation. This is where the notion of *viscosity solutions* comes into the picture.

Remark 2.2. Since Theorem 1.6 shows that u is Lipschitz continuous, Rademacher's theorem [EG15] gives that u is differentiable almost everywhere. One natural suggestion would be to look for almost everywhere solutions. The only problem with this approach is that uniqueness fails; we shall soon see an example of this.

But before we define viscosity solutions and motivate their uniqueness, let us first generalise our set-up from *Hamilton-Jacobi-Bellman equations* to *Hamilton-Jacobi equations*.

2.1 *Hamilton-Jacobi Equations* (A generalisation)

Throughout this chapter we use the notation u_t and Du introduced in Chapter 1 (Du is the gradient in the space variable only). Let $p = Du$ and recall (1.15). Define the *Hamiltonian* $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$H(p, x) := \inf_{a \in A} \Big\{ f(x, a) \cdot p + r(x, a) \Big\}. \tag{2.1}$$

Then (1.15) can be written as

$$u_t + H(Du, x) = 0, \tag{2.2}$$

which we call the *Hamilton–Jacobi equation*.

In what follows we work with a more general class of Hamiltonians.

Definition 2.3 (HAMILTONIAN [Eva10])**.** A function $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is called a *Hamiltonian* if there exists a constant $C > 0$ such that for all $p, q, x, y \in \mathbb{R}^n$,

$$|H(p, x) - H(q, x)| \le C|p - q|, \tag{2.3}$$
$$|H(p, x) - H(p, y)| \le C|x - y|\,(1 + |p|). \tag{2.4}$$

We now check that H defined in (2.1) is a Hamiltonian in the sense of Definition 2.3. For this we need the following elementary fact:

Lemma 2.4. *Let A be a nonempty set and let $\varphi, \psi : A \to \mathbb{R}$ be bounded functions. Then*

$$\Big| \inf_A \varphi - \inf_A \psi \Big| \le \sup_A |\varphi - \psi|.$$

Proof. For every $a \in A$, $\varphi(a) \le \psi(a) + \sup_A |\varphi - \psi|$. Taking the infimum over $a \in A$ gives $\inf_A \varphi \le \inf_A \psi + \sup_A |\varphi - \psi|$. Exchanging the roles of φ and ψ yields the reverse inequality. □

Proposition 2.5. *Assume (1.1) and (1.4) hold. Then H defined in (2.1) is a Hamiltonian in the sense of Definition 2.3.*

Proof. For fixed p, x, the map $a \mapsto f(x, a) \cdot p + r(x, a)$ is bounded by $M|p| + C$, so H is finite. We show that (2.3) and (2.4) hold with the constant $\max\{M, L, C\}$.

Proof of (2.3). Fix x and set $\varphi(a) = f(x, a) \cdot p + r(x, a)$, $\psi(a) = f(x, a) \cdot q + r(x, a)$. By the Cauchy–Schwarz inequality and the bound on f,

$$|\varphi(a) - \psi(a)| = |f(x, a) \cdot (p - q)| \le M|p - q| \quad \text{for all } a \in A,$$

so Lemma 2.4 gives (2.3).

Proof of (2.4). Fix p and set $\varphi(a) = f(x,a)\cdot p + r(x,a)$, $\psi(a) = f(y,a)\cdot p + r(y,a)$. Then

$$\begin{aligned}|\varphi(a) - \psi(a)| &\le |f(x,a) - f(y,a)|\,|p| + |r(x,a) - r(y,a)| \\ &\le L|x-y|\,|p| + C|x-y| \le \max\{L,C\}\,|x-y|(1+|p|)\end{aligned}$$

for all $a \in A$, so Lemma 2.4 gives (2.4). □

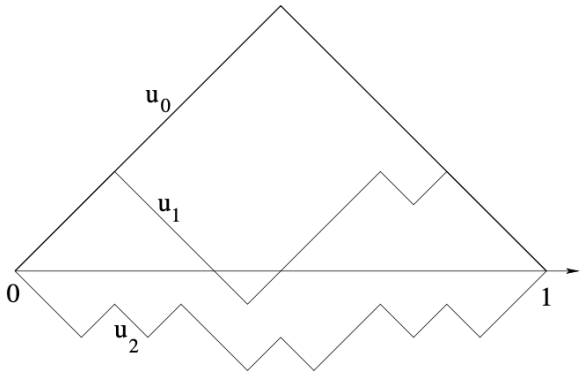
2.2 Motivation for Viscosity Solutions

As discussed in Remark 2.2, almost everywhere solutions could be a candidate for solving *Hamilton-Jacobi equations* in some weak sense, but uniqueness fails. Here is an example that demonstrates the failure of uniqueness for almost everywhere solutions:

Example 2.6. Consider the boundary value problem on the unit interval

$$|u_x| - 1 = 0, \quad x \in [0,1], \qquad u(0) = u(1) = 0. \tag{2.5}$$

This equation is famously known as the *Eikonal equation*. By a generalized solution we mean a Lipschitz continuous function $u : [0,1] \to \mathbb{R}$ that satisfies $|u'(x)| = 1$ at every point where u is differentiable (that is, almost everywhere) and the boundary conditions. Such a function is necessarily piecewise linear with slopes ± 1. Problem (2.5) has infinitely many generalized solutions; consider the figure below [Bre].



$u_0(x) = \min\{x,\, 1-x\}$ is a solution: it has slope $+1$ on $(0, \frac{1}{2})$ and slope -1 on $(\frac{1}{2}, 1)$, and it is not differentiable only at $x = \frac{1}{2}$. More generally, for $n \in \mathbb{N}$ let $u^{(n)}(x)$ be the distance from x to the nearest point of the set $\{0, \frac{1}{n}, \frac{2}{n}, \dots, 1\}$. Its graph is a sawtooth with n teeth, with slope ± 1 away from the corners, so $u^{(n)}$ is a generalized solution for every n; so is $-u^{(n)}$. Further solutions are obtained by changing the direction of the slope at different points, as in the functions u_1 and u_2 of the figure. Hence generalized solutions are not unique.

In Section 2.6 we show that u_0 is a viscosity solution of (2.5), while $u^{(2)}$ is not.

2.3 Definitions and compatibility

Strategy

The optimal control problem of Chapter 1 leads to the *terminal*-value problem (2.7): the cost is prescribed at the final time $t = T$ and the value function is determined backward in time. The theory of viscosity solutions, however, is usually developed for *initial*-value problems, where data is prescribed at $t = 0$ and the solution evolves forward in time [Eva10]. We therefore proceed in three steps.

1. We state the definition of a viscosity solution for the initial-value problem (2.6) with a general Hamiltonian G (Definition 2.7). The theory developed from this definition depends on G only through the estimates (2.3)–(2.4).

2. We define viscosity solutions of the terminal-value problem (2.7) (Definition 2.8) and show in Proposition 2.10 that u is a viscosity solution of (2.7) if and only if $w(x,t) = u(x, T-t)$ is a viscosity solution of an initial-value problem with Hamiltonian $-H$.

3. Since $-H$ satisfies (2.3)–(2.4) whenever H does (Lemma 2.9), every result for initial-value problems applies to w, and hence, through the correspondence in step 2, to u.

Consequently, for this section we work with the initial-value problem; the results transfer to the control problem by time reversal. Only the sign of the Hamiltonian and the direction of the inequalities change, not the theory.

Initial-value problem

Let G be a Hamiltonian in the sense of Definition 2.3, let $T > 0$, and let $g : \mathbb{R}^n \to \mathbb{R}$ be bounded and Lipschitz. Consider

$$\begin{cases} w_t + G(Dw, x) = 0 & \text{in } \mathbb{R}^n \times (0,T), \\ w = g & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases} \tag{2.6}$$

Definition 2.7 (VISCOSITY SOLUTION, INITIAL VALUE PROBLEM [Eva10])**.** A bounded, uniformly continuous function $w : \mathbb{R}^n \times [0,T] \to \mathbb{R}$ is a *viscosity solution* of (2.6) if

(i) $w = g$ on $\mathbb{R}^n \times \{t = 0\}$, and

(ii) for every $v \in C^\infty(\mathbb{R}^n \times (0,T))$:

- if $w - v$ has a local maximum at $(x_0, t_0) \in \mathbb{R}^n \times (0,T)$, then

$$v_t(x_0,t_0) + G(Dv(x_0,t_0), x_0) \le 0;$$

- if $w - v$ has a local minimum at $(x_0, t_0) \in \mathbb{R}^n \times (0,T)$, then

$$v_t(x_0,t_0) + G(Dv(x_0,t_0), x_0) \ge 0.$$

Terminal-value problem

We are interested in the terminal-value problem, since the optimal control problem leads to

$$\begin{cases} u_t + H(Du, x) = 0 & \text{in } \mathbb{R}^n \times (0,T), \\ u = g & \text{on } \mathbb{R}^n \times \{t = T\}, \end{cases} \tag{2.7}$$

with H as in (2.1). Since the data is given at $t = T$, the Definition 2.7 does not apply directly.

Definition 2.8 (VISCOSITY SOLUTION, TERMINAL VALUE PROBLEM [Eva10])**.** A bounded, uniformly continuous function $u : \mathbb{R}^n \times [0,T] \to \mathbb{R}$ is a *viscosity solution* of (2.7) if

(a) $u = g$ on $\mathbb{R}^n \times \{t = T\}$, and

(b) for every $v \in C^\infty(\mathbb{R}^n \times (0,T))$:

- if $u - v$ has a local maximum at $(x_0, t_0) \in \mathbb{R}^n \times (0,T)$, then
$$v_t(x_0,t_0) + H(Dv(x_0,t_0), x_0) \geq 0;$$
- if $u - v$ has a local minimum at $(x_0, t_0) \in \mathbb{R}^n \times (0,T)$, then
$$v_t(x_0,t_0) + H(Dv(x_0,t_0), x_0) \leq 0.$$

Note that the inequalities are reversed compared with Definition 2.7. The next proposition explains why.

Lemma 2.9. *If H is a Hamiltonian in the sense of Definition 2.3, so is $-H$.*

Proof. $|(-H)(p,x) - (-H)(q,x)| = |H(p,x) - H(q,x)|$, and likewise in x, so (2.3) and (2.4) hold with the same C. □

Proposition 2.10 (Time reversal)**.** *Let $u : \mathbb{R}^n \times [0,T] \to \mathbb{R}$ be bounded and uniformly continuous, and set*

$$w(x,t) := u(x, T-t), \qquad (x,t) \in \mathbb{R}^n \times [0,T].$$

Then u is a viscosity solution of the terminal-value problem (2.7) if and only if w is a viscosity solution of the initial-value problem

$$\begin{cases} w_t - H(Dw, x) = 0 & \text{in } \mathbb{R}^n \times (0,T), \\ w = g & \text{on } \mathbb{R}^n \times \{t = 0\}, \end{cases} \tag{2.8}$$

that is, of (2.6) with $G = -H$ (a Hamiltonian by Lemma 2.9).

Proof. w is bounded and uniformly continuous exactly when u is, and $w(\cdot,0) = u(\cdot,T)$, so $u = g$ at $t = T$ if and only if $w = g$ at $t = 0$.

For $v \in C^\infty(\mathbb{R}^n \times (0,T))$ define $\tilde{v}(x,s) := v(x, T-s)$. The map $v \mapsto \tilde{v}$ is a bijection of $C^\infty(\mathbb{R}^n \times (0,T))$ onto itself. Since $(x,t) \mapsto (x, T-t)$ is a homeomorphism of $\mathbb{R}^n \times (0,T)$,

$$w - v \text{ has a local max (min) at } (x_0,t_0) \iff u - \tilde{v} \text{ has a local max (min) at } (x_0, T-t_0).$$

Moreover, by the chain rule,

$$\tilde{v}_s(x_0, T-t_0) = -v_t(x_0,t_0), \qquad D\tilde{v}(x_0,T-t_0) = Dv(x_0,t_0). \tag{2.9}$$

Suppose u satisfies (b) and let $w-v$ have a local maximum at (x_0,t_0). Then $u-\tilde{v}$ has a local maximum at $(x_0,T-t_0)$, so

$$\tilde{v}_s(x_0,T-t_0) + H(D\tilde{v}(x_0,T-t_0),x_0) \geq 0.$$

By (2.9) this reads $-v_t(x_0,t_0) + H(Dv(x_0,t_0),x_0) \geq 0$, i.e.

$$v_t(x_0,t_0) + (-H)(Dv(x_0,t_0),x_0) \leq 0,$$

which is the sub-solution condition of Definition 2.7 for $G=-H$. The case of a local minimum is identical, with all inequalities reversed. Hence w is a viscosity solution of (2.8).

Conversely, if w is a viscosity solution of (2.8), apply the same argument backwards, using that $v \mapsto \tilde{v}$ is a bijection and that (2.9) is an equivalence. Thus u satisfies (b). □

Remark 2.11. Proposition 2.10 says that the reversed inequalities in Definition 2.8 are not an extra assumption. They are the usual viscosity inequalities for w, written in the original time variable. Thus the two definitions describe the same notion of solution, seen from the two ends of the time interval, and no new theory is needed for the terminal-value problem.

2.4 Consistency of viscosity solutions

Throughout, H is a Hamiltonian in the sense of Definition 2.3. In particular H is jointly continuous, because

$$|H(p,x)-H(q,y)| \leq |H(p,x)-H(q,x)|+|H(q,x)-H(q,y)| \leq C|p-q|+C|x-y|(1+|q|).$$

We consider the initial-value problem

$$\begin{cases} u_t + H(Du,x) = 0 & \text{in } \mathbb{R}^n \times (0,\infty), \\ u = g & \text{on } \mathbb{R}^n \times \{t=0\}, \end{cases} \tag{2.10}$$

and viscosity solutions as in Definition 2.7, with test functions $v \in C^\infty(\mathbb{R}^n \times (0,\infty))$. Before developing the theory further, we check that this notion is consistent with classical solutions.

Classical solutions are viscosity solutions

Proposition 2.12. *Let $u \in C^1(\mathbb{R}^n \times [0,\infty))$ be bounded and uniformly continuous, and let u solve (2.10) in the classical sense. Then u is a viscosity solution of (2.10).*

Proof. The initial condition $u=g$ holds because u is a classical solution, so we only have to verify condition (ii) of Definition 2.7.

Let $v \in C^\infty(\mathbb{R}^n \times (0,\infty))$ and suppose $u - v$ has a local maximum at $(x_0, t_0) \in \mathbb{R}^n \times (0,\infty)$. The point (x_0, t_0) is an interior point of the open set $\mathbb{R}^n \times (0,\infty)$, and $u - v$ is differentiable there, so all first partial derivatives of $u - v$ vanish at (x_0, t_0). That is,

$$Du(x_0, t_0) = Dv(x_0, t_0), \qquad u_t(x_0, t_0) = v_t(x_0, t_0).$$

Consequently

$$v_t(x_0, t_0) + H(Dv(x_0, t_0), x_0) = u_t(x_0, t_0) + H(Du(x_0, t_0), x_0) = 0,$$

since u solves the equation at (x_0, t_0). In particular $v_t + H(Dv, x_0) \leq 0$ at (x_0, t_0), as required.

If $u - v$ has a local minimum at (x_0, t_0), the same argument gives

$$v_t(x_0, t_0) + H(Dv(x_0, t_0), x_0) = 0 \geq 0. \qquad \square$$

Differentiable viscosity solutions satisfy the equation

The converse is that a viscosity solution which is differentiable at a point satisfies the PDE there. We need a calculus fact: a function that is differentiable at a point can be touched from above by a C^1 function there.

Lemma 2.13 (TOUCHING BY A C^1 FUNCTION [Eva10])**.** *Let $u : \mathbb{R}^m \to \mathbb{R}$ be continuous and differentiable at the point x_0. Then there exists $v \in C^1(\mathbb{R}^m)$ such that*

$$u(x_0) = v(x_0) \tag{2.11}$$

and

$$u - v \text{ has a strict local maximum at } x_0. \tag{2.12}$$

Proof. Step 1: Reduction to $x_0 = 0$, $u(0) = 0$, $Du(0) = 0$. Define

$$\tilde{u}(x) := u(x + x_0) - u(x_0) - Du(x_0) \cdot x.$$

Then $\tilde{u}$ is continuous, differentiable at 0, and

$$\tilde{u}(0) = 0, \qquad D\tilde{u}(0) = 0. \tag{2.13}$$

Suppose we have found $\tilde{v} \in C^1(\mathbb{R}^m)$ with $\tilde{v}(0) = 0$ such that $\tilde{u} - \tilde{v}$ has a strict local maximum at 0. Put

$$v(x) := \tilde{v}(x - x_0) + u(x_0) + Du(x_0) \cdot (x - x_0).$$

Then $v \in C^1(\mathbb{R}^m)$, $v(x_0) = u(x_0)$, and

$$u(x) - v(x) = \tilde{u}(x - x_0) - \tilde{v}(x - x_0),$$

so $u - v$ has a strict local maximum at x_0. Hence it suffices to treat $\tilde{u}$, and from now on we assume (2.13) for u with $x_0 = 0$.

Step 2: A modulus for u. Define $\rho_1 : \mathbb{R}^m \to \mathbb{R}$ by $\rho_1(x) := u(x)/|x|$ for $x \neq 0$ and $\rho_1(0) := 0$. Then ρ_1 is continuous at every $x \neq 0$ because u is continuous, and continuous at 0 because $u(0) = 0$, $Du(0) = 0$ means exactly $u(x)/|x| \to 0$ as $x \to 0$. Thus

$$u(x) = |x|\rho_1(x), \qquad \rho_1 \text{ continuous, } \rho_1(0) = 0. \tag{2.14}$$

Set

$$\rho_2(r) := \max_{|x|\le r} |\rho_1(x)| \qquad (r \ge 0).$$

The maximum exists since ρ_1 is continuous and the closed ball is compact. We claim

$$\rho_2 : [0,\infty) \to [0,\infty) \text{ is continuous, nondecreasing, } \rho_2(0) = 0. \tag{2.15}$$

Clearly $\rho_2(0) = |\rho_1(0)| = 0$, and ρ_2 is nondecreasing since the balls increase with r. For continuity, fix $R > 0$ and let ω be a modulus of continuity of ρ_1 on the compact set $\overline{B}(0,R)$. Let $0 \le r < s \le R$ and $|x| \le s$. The point $x' := \frac{r}{s}x$ satisfies $|x'| \le r$ and $|x - x'| = (1 - \frac{r}{s})|x| \le s - r$. Hence $|\rho_1(x)| \le |\rho_1(x')| + \omega(s-r) \le \rho_2(r) + \omega(s-r)$. Taking the maximum over $|x| \le s$,

$$0 \le \rho_2(s) - \rho_2(r) \le \omega(s-r),$$

which gives continuity on $[0,R]$ for every R.

Step 3: Definition of v and its regularity. Define

$$v(x) := \int_{|x|}^{2|x|} \rho_2(r)\,dr + |x|^2 \qquad (x \in \mathbb{R}^m).$$

Since $\rho_2 \ge 0$ is nondecreasing, $0 \le \int_{|x|}^{2|x|} \rho_2\,dr \le |x|\rho_2(2|x|)$, so

$$|v(x)| \le |x|\rho_2(2|x|) + |x|^2. \tag{2.16}$$

Dividing by $|x|$ and letting $x \to 0$ (using $\rho_2(0) = 0$ and continuity of ρ_2) shows $v(x)/|x| \to 0$, that is,

$$v(0) = 0, \qquad Dv(0) = 0. \tag{2.17}$$

For $x \neq 0$, let $\Phi(s) := \int_s^{2s} \rho_2(r)\,dr$ for $s > 0$. Since ρ_2 is continuous, the fundamental theorem of calculus gives $\Phi \in C^1((0,\infty))$ with $\Phi'(s) = 2\rho_2(2s) - \rho_2(s)$. As $x \mapsto |x|$ is smooth on $\mathbb{R}^m \setminus \{0\}$ with $D|x| = x/|x|$, the chain rule gives

$$Dv(x) = \big(2\rho_2(2|x|) - \rho_2(|x|)\big)\frac{x}{|x|} + 2x \qquad (x \neq 0). \tag{2.18}$$

This is continuous on $\mathbb{R}^m \setminus \{0\}$. It is also continuous at 0, because $\big|Dv(x)\big| \le 2\rho_2(2|x|) + \rho_2(|x|) + 2|x| \to 0 = Dv(0)$ as $x \to 0$. Hence $v \in C^1(\mathbb{R}^m)$.

Step 4: $u - v$ has a strict maximum at 0. Let $x \neq 0$. By (2.14) and the definition of ρ_2, $u(x) = |x|\rho_1(x) \le |x|\rho_2(|x|)$. Since ρ_2 is nondecreasing, $\rho_2(r) \ge \rho_2(|x|)$ for $r \in [|x|, 2|x|]$, so $\int_{|x|}^{2|x|} \rho_2\,dr \ge |x|\rho_2(|x|)$. Therefore

$$u(x) - v(x) \le |x|\rho_2(|x|) - \int_{|x|}^{2|x|} \rho_2(r)\,dr - |x|^2 \le -|x|^2 < 0 = u(0) - v(0).$$

Thus $u - v$ has a strict (indeed global) maximum at 0, and $v(0) = u(0)$. □

Theorem 2.14 (CONSISTENCY OF VISCOSITY SOLUTION [Eva10])**.** *Let u be a viscosity solution of (2.10), and suppose u is differentiable at some point $(x_0, t_0) \in \mathbb{R}^n \times (0,\infty)$. Then*

$$u_t(x_0,t_0) + H(Du(x_0,t_0), x_0) = 0.$$

Proof. Step 1: Touching from above. Lemma 2.13 is stated for functions on all of $\mathbb{R}^{n+1}$, whereas u is only defined for $t \geq 0$. Since $t_0 > 0$ this is harmless: the function $\bar{u}(x,t) := u(x, \max\{t, t_0/2\})$ is continuous on $\mathbb{R}^{n+1}$ and coincides with u on $\{t \geq t_0/2\}$, a neighbourhood of (x_0,t_0), so $\bar{u}$ is differentiable at (x_0,t_0) with the same derivatives as u. Applying Lemma 2.13 with $m = n+1$ to $\bar{u}$ at (x_0,t_0) gives $v \in C^1(\mathbb{R}^{n+1})$ such that

$$u - v \text{ has a strict local maximum at } (x_0,t_0). \tag{2.19}$$

Fix $r > 0$ so small that $B := \overline{B}((x_0,t_0),r) \subset \mathbb{R}^n \times (0,\infty)$ and

$$(u-v)(x,t) < (u-v)(x_0,t_0) \quad \text{for all } (x,t) \in B \setminus \{(x_0,t_0)\}.$$

Step 2: Mollification. The function v is only C^1, whereas Definition 2.7 requires C^∞ test functions. Let η_ε be the standard mollifier in the $n+1$ variables (x,t) and set $v^\varepsilon := \eta_\varepsilon * v \in C^\infty(\mathbb{R}^{n+1})$. Since v and its first derivatives are continuous,

$$v^\varepsilon \to v, \quad Dv^\varepsilon \to Dv, \quad v_t^\varepsilon \to v_t \quad \text{uniformly on } B \text{ as } \varepsilon \to 0, \tag{2.20}$$

(using $D(\eta_\varepsilon * v) = \eta_\varepsilon * Dv$).

Step 3: Maxima of $u - v^\varepsilon$. The continuous function $u - v^\varepsilon$ attains its maximum over the compact set B at some point $(x_\varepsilon, t_\varepsilon) \in B$. We claim

$$(x_\varepsilon,t_\varepsilon) \to (x_0,t_0) \quad \text{as } \varepsilon \to 0. \tag{2.21}$$

Let $\delta_\varepsilon := \sup_B |v^\varepsilon - v|$, so $\delta_\varepsilon \to 0$ by (2.20). By maximality of $(x_\varepsilon,t_\varepsilon)$,

$$(u-v)(x_\varepsilon,t_\varepsilon) \geq (u-v^\varepsilon)(x_\varepsilon,t_\varepsilon) - \delta_\varepsilon \geq (u-v^\varepsilon)(x_0,t_0) - \delta_\varepsilon \geq (u-v)(x_0,t_0) - 2\delta_\varepsilon.$$

Let $\varepsilon_k \to 0$ be any sequence. By compactness of B, a subsequence converges to some $(x^*,t^*) \in B$, and passing to the limit in the display (using continuity of $u-v$) gives $(u-v)(x^*,t^*) \geq (u-v)(x_0,t_0)$. By the strictness in Step 1, $(x^*,t^*) = (x_0,t_0)$. Since every sequence has a subsequence converging to (x_0,t_0), (2.21) follows.

In particular, for ε small, $(x_\varepsilon,t_\varepsilon)$ lies in the interior of B, so it is a local maximum of $u - v^\varepsilon$ in $\mathbb{R}^n \times (0,\infty)$.

Step 4: Using the definition. Since v^ε is smooth, Definition 2.7(ii) gives

$$v_t^\varepsilon(x_\varepsilon,t_\varepsilon) + H\Big(Dv^\varepsilon(x_\varepsilon,t_\varepsilon), x_\varepsilon\Big) \leq 0. \tag{2.22}$$

We let $\varepsilon \to 0$. By (2.20) and (2.21),

$$|v_t^\varepsilon(x_\varepsilon,t_\varepsilon) - v_t(x_0,t_0)| \leq \sup_B |v_t^\varepsilon - v_t| + |v_t(x_\varepsilon,t_\varepsilon) - v_t(x_0,t_0)| \to 0,$$

and in the same way $Dv^\varepsilon(x_\varepsilon,t_\varepsilon) \to Dv(x_0,t_0)$. Since H is continuous, (2.22) yields

$$v_t(x_0,t_0) + H(Dv(x_0,t_0),x_0) \leq 0. \tag{2.23}$$

Step 5: Identifying the derivatives. By (2.19), $u - v$ has a local maximum at the interior point (x_0,t_0), and $u - v$ is differentiable there (as u is, and v is C^1). Hence the gradient of $u - v$ vanishes at (x_0,t_0):

$$Du(x_0,t_0) = Dv(x_0,t_0), \qquad u_t(x_0,t_0) = v_t(x_0,t_0).$$

Substituting into (2.23),

$$u_t(x_0,t_0) + H(Du(x_0,t_0),x_0) \le 0. \tag{2.24}$$

Step 6: The reverse inequality. The function $-u$ is continuous and differentiable at (x_0,t_0). Applying Lemma 2.13 to $-\bar{u}$ yields $\tilde{v} \in C^1(\mathbb{R}^{n+1})$ such that $(-u)-\tilde{v}$ has a strict local maximum at (x_0,t_0). Setting $v := -\tilde{v}$, we get that

$$u - v \text{ has a strict local minimum at } (x_0,t_0).$$

Now repeat Steps 2–5 with "minimum" in place of "maximum": mollify v, choose points $(x_\varepsilon,t_\varepsilon)$ where $u-v^\varepsilon$ attains its *minimum* over B (they converge to (x_0,t_0) by the same argument with all inequalities reversed), apply the second condition in Definition 2.7(ii) to get $v_t^\varepsilon(x_\varepsilon,t_\varepsilon) + H(Dv^\varepsilon(x_\varepsilon,t_\varepsilon),x_\varepsilon) \ge 0$, let $\varepsilon \to 0$, and use $Du = Dv$, $u_t = v_t$ at (x_0,t_0). This gives

$$u_t(x_0,t_0) + H(Du(x_0,t_0),x_0) \ge 0. \tag{2.25}$$

Combining (2.24) and (2.25) proves the theorem. □

Corollary 2.15. *If $u \in C^1(\mathbb{R}^n \times (0,\infty))$ is a viscosity solution of (2.10), then $u_t + H(Du,x) = 0$ at every point of $\mathbb{R}^n \times (0,\infty)$, i.e. u is a classical solution. Together with Proposition 2.12, classical and viscosity solutions coincide within the class of bounded, uniformly continuous C^1 functions.*

Remark 2.16. Everything above holds for any Hamiltonian G satisfying (2.3)–(2.4), in particular for $G = -H$. By Proposition 2.10, the consistency result therefore also holds for the terminal-value problem (2.7), with the inequalities in Definition 2.8.

2.5 Uniqueness of Viscosity Solutions

We now show that the initial-value problem (2.6) has at most one viscosity solution. The proof is based on the idea of "doubling the number of variables" [Eva10].

Theorem 2.17 (UNIQUENESS OF VISCOSITY SOLUTIONS [Eva10])**.** *Let G be a Hamiltonian in the sense of Definition 2.3, and let g be bounded and Lipschitz. Then there exists at most one viscosity solution of (2.6).*

Proof. Let u and $\tilde{u}$ be two viscosity solutions of (2.6); in particular both are bounded and uniformly continuous on $\mathbb{R}^n \times [0,T]$ and $u(\cdot,0) = \tilde{u}(\cdot,0) = g$. It suffices to show $u \le \tilde{u}$; exchanging the roles of u and $\tilde{u}$ then gives $\tilde{u} \le u$. Suppose, to the contrary, that

$$\sup_{\mathbb{R}^n \times [0,T]} (u - \tilde{u}) =: \sigma > 0. \tag{2.26}$$

Let ω and $\tilde{\omega}$ be the (nondecreasing) moduli of continuity of u and $\tilde{u}$:

$$|u(x,t)-u(y,s)| \le \omega(|x-y|+|t-s|), \qquad |\tilde{u}(x,t)-\tilde{u}(y,s)| \le \tilde{\omega}(|x-y|+|t-s|),$$

with $\omega(r),\tilde{\omega}(r) \to 0$ as $r \to 0$. Let $K_0 := \sup|u| + \sup|\tilde{u}|$.

Step 1: The doubled function and its maximum point. Since $u-\tilde{u}$ is continuous, there is (x_1,t_1) with $t_1 < T$ and $(u-\tilde{u})(x_1,t_1) \ge \frac{3}{4}\sigma$. Fix $\lambda,\eta \in (0,1)$ so small that

$$2\lambda T \le \frac{\sigma}{12}, \qquad \frac{2\eta}{T-t_1} \le \frac{\sigma}{12}. \tag{2.27}$$

For $0 < \varepsilon < 1$ and $x, y \in \mathbb{R}^n$, $t, s \in [0, T)$ set

$$\begin{aligned}\Phi(x, y, t, s) := u(x, t) - \tilde{u}(y, s) - \lambda(t + s) - \frac{\eta}{T - t} - \frac{\eta}{T - s} \\ - \frac{1}{\varepsilon^2}\Big(|x - y|^2 + (t - s)^2\Big) - \varepsilon\Big(|x|^2 + |y|^2\Big).\end{aligned} \tag{2.28}$$

The terms with η keep the maximum point away from $t = T$, $s = T$, where Definition 2.7 gives no information because test functions are only defined on $\mathbb{R}^n \times (0, T)$. The function Φ is continuous, and since $u, \tilde{u}$ are bounded, $\Phi \to -\infty$ as $|x| + |y| \to \infty$, or $t \to T$, or $s \to T$. Hence the set $\{\Phi \geq \Phi(0, 0, 0, 0)\}$ is compact and there is a point $(x_0, y_0, t_0, s_0) \in \mathbb{R}^{2n} \times [0, T)^2$ (depending on ε) with

$$\Phi(x_0, y_0, t_0, s_0) = \max \Phi. \tag{2.29}$$

Step 2: A lower bound for Φ and first estimates. Choose $\varepsilon_0 \in (0, 1)$ with $2\varepsilon_0|x_1|^2 \leq \sigma/12$. For $\varepsilon < \varepsilon_0$, by (2.27),

$$\begin{aligned}\Phi(x_0, y_0, t_0, s_0) &\geq \Phi(x_1, x_1, t_1, t_1) \\ &= (u - \tilde{u})(x_1, t_1) - 2\lambda t_1 - \frac{2\eta}{T - t_1} - 2\varepsilon|x_1|^2 \\ &\geq \frac{3}{4}\sigma - \frac{3}{12}\sigma = \frac{\sigma}{2}.\end{aligned} \tag{2.30}$$

In addition, $\Phi(x_0, y_0, t_0, s_0) \geq \Phi(0, 0, 0, 0) = u(0, 0) - \tilde{u}(0, 0) - 2\eta/T$, hence, with $K := 2K_0 + 2/T$,

$$\begin{aligned}\lambda(t_0 + s_0) + \frac{\eta}{T - t_0} + \frac{\eta}{T - s_0} + \frac{1}{\varepsilon^2}\Big(|x_0 - y_0|^2 + (t_0 - s_0)^2\Big) + \varepsilon\Big(|x_0|^2 + |y_0|^2\Big) \\ \leq u(x_0, t_0) - \tilde{u}(y_0, s_0) - u(0, 0) + \tilde{u}(0, 0) + \frac{2\eta}{T} \leq K.\end{aligned} \tag{2.31}$$

Since u and $\tilde{u}$ are bounded, every term on the left is at most K. In particular

$$|x_0 - y_0|,\ |t_0 - s_0| \leq \sqrt{K}\,\varepsilon = O(\varepsilon) \quad \text{as } \varepsilon \to 0. \tag{2.32}$$

Furthermore $\varepsilon(|x_0|^2 + |y_0|^2) \leq K$, and consequently

$$\begin{aligned}\varepsilon(|x_0| + |y_0|) = \varepsilon^{1/4}\,\varepsilon^{3/4}(|x_0| + |y_0|) &\leq \frac{1}{2}\varepsilon^{1/2} + \frac{1}{2}\varepsilon^{3/2}(|x_0| + |y_0|)^2 \\ &\leq \frac{1}{2}\varepsilon^{1/2} + \varepsilon^{1/2} \cdot \varepsilon(|x_0|^2 + |y_0|^2) \leq \Big(\frac{1}{2} + K\Big)\varepsilon^{1/2}.\end{aligned}$$

Thus

$$\varepsilon(|x_0| + |y_0|) = O(\varepsilon^{1/2}) \quad \text{as } \varepsilon \to 0. \tag{2.33}$$

Step 3: A sharper estimate. Since $\Phi(x_0, y_0, t_0, s_0) \geq \Phi(x_0, x_0, t_0, t_0)$, we have

$$\begin{aligned}u(x_0, t_0) - \tilde{u}(y_0, s_0) - \lambda(t_0 + s_0) - \frac{\eta}{T - t_0} - \frac{\eta}{T - s_0} \\ - \frac{1}{\varepsilon^2}\Big(|x_0 - y_0|^2 + (t_0 - s_0)^2\Big) - \varepsilon(|x_0|^2 + |y_0|^2) \\ \geq u(x_0, t_0) - \tilde{u}(x_0, t_0) - 2\lambda t_0 - \frac{2\eta}{T - t_0} - 2\varepsilon|x_0|^2.\end{aligned}$$

Hence, using $|x_0|^2 - |y_0|^2 = (x_0+y_0)\cdot(x_0-y_0)$ and $\frac{1}{T-t_0} - \frac{1}{T-s_0} = \frac{t_0-s_0}{(T-t_0)(T-s_0)}$,

$$\begin{aligned}\frac{1}{\varepsilon^2}\Big(|x_0-y_0|^2 + (t_0-s_0)^2\Big) \le \tilde{u}(x_0,t_0) - \tilde{u}(y_0,s_0) + \lambda(t_0-s_0)\\ + \varepsilon(x_0+y_0)\cdot(x_0-y_0) + \frac{\eta\,(t_0-s_0)}{(T-t_0)(T-s_0)}.\end{aligned} \tag{2.34}$$

We bound the terms on the right as $\varepsilon \to 0$. By (2.32) and the uniform continuity of $\tilde{u}$, the first is at most $\tilde{\omega}(2\sqrt{K}\varepsilon) \to 0$. The second is $O(\varepsilon)$ by (2.32). The third is at most $\varepsilon(|x_0|+|y_0|)\,|x_0-y_0| = O(\varepsilon^{1/2})O(\varepsilon) \to 0$ by (2.33). For the last, (2.31) gives $\eta/(T-t_0) \le K$ and $\eta/(T-s_0) \le K$, so $\frac{\eta}{(T-t_0)(T-s_0)} \le \frac{K^2}{\eta}$ and the term is at most $\frac{K^2}{\eta}\sqrt{K}\,\varepsilon \to 0$ (η is fixed). So the right-hand side of (2.34) tends to 0 as $\varepsilon \to 0$, that is,

$$|x_0-y_0|,\ |t_0-s_0| = o(\varepsilon) \quad \text{as } \varepsilon \to 0. \tag{2.35}$$

Step 4: The maximum point is away from $t=0$. All the subtracted terms in Φ are non-negative, so (2.30) gives

$$\frac{\sigma}{2} \le u(x_0,t_0) - \tilde{u}(y_0,s_0).$$

Writing $u(x_0,t_0) - \tilde{u}(y_0,s_0)$ as a telescoping sum and using $u(\cdot,0) = \tilde{u}(\cdot,0)$,

$$\begin{aligned}u(x_0,t_0) - \tilde{u}(y_0,s_0) &= u(x_0,t_0) - u(x_0,0) + \underbrace{u(x_0,0) - \tilde{u}(x_0,0)}_{=0}\\ &\quad + \tilde{u}(x_0,0) - \tilde{u}(x_0,t_0) + \tilde{u}(x_0,t_0) - \tilde{u}(y_0,s_0)\\ &\le \omega(t_0) + \tilde{\omega}(t_0) + \tilde{\omega}(|x_0-y_0| + |t_0-s_0|).\end{aligned}$$

By (2.32), we can take ε so small that the last term is at most $\sigma/4$; then $\frac{\sigma}{4} \le \omega(t_0) + \tilde{\omega}(t_0)$. Choose $\mu > 0$ with $\omega(\mu) + \tilde{\omega}(\mu) < \sigma/4$. Since $\omega, \tilde{\omega}$ are nondecreasing, this forces $t_0 > \mu$. By (2.32) we also get $s_0 \ge t_0 - \sqrt{K}\varepsilon > \mu/2$ for ε small. Together with $t_0, s_0 < T$:

$$t_0,\ s_0 \in (\mu/2, T). \tag{2.36}$$

Step 5: Using the definition of viscosity solution. By (2.29), the map $(x,t) \mapsto \Phi(x,y_0,t,s_0)$ has a maximum at (x_0,t_0). By (2.28) this means that $u - v$ has a maximum at (x_0,t_0), where

$$v(x,t) := \tilde{u}(y_0,s_0) + \lambda(t+s_0) + \frac{\eta}{T-t} + \frac{\eta}{T-s_0} + \frac{1}{\varepsilon^2}\Big(|x-y_0|^2 + (t-s_0)^2\Big) + \varepsilon\Big(|x|^2 + |y_0|^2\Big).$$

The function v belongs to $C^\infty(\mathbb{R}^n \times (0,T))$ and, by (2.36), $(x_0,t_0) \in \mathbb{R}^n \times (0,T)$. Since u is a viscosity solution, Definition 2.7 gives $v_t(x_0,t_0) + G(Dv(x_0,t_0),x_0) \le 0$, that is,

$$\lambda + \frac{\eta}{(T-t_0)^2} + \frac{2(t_0-s_0)}{\varepsilon^2} + G\Big(\frac{2}{\varepsilon^2}(x_0-y_0) + 2\varepsilon x_0,\ x_0\Big) \le 0. \tag{2.37}$$

Similarly, the map $(y,s) \mapsto -\Phi(x_0,y,t_0,s)$ has a minimum at (y_0,s_0), so $\tilde{u} - \tilde{v}$ has a minimum at (y_0,s_0), where

$$\tilde{v}(y,s) := u(x_0,t_0) - \lambda(t_0+s) - \frac{\eta}{T-t_0} - \frac{\eta}{T-s} - \frac{1}{\varepsilon^2}\Big(|x_0-y|^2 + (t_0-s)^2\Big) - \varepsilon\Big(|x_0|^2 + |y|^2\Big).$$

As $\tilde{u}$ is a viscosity solution, $\tilde{v}_s(y_0, s_0) + G(D\tilde{v}(y_0, s_0), y_0) \geq 0$, that is,

$$-\lambda - \frac{\eta}{(T-s_0)^2} + \frac{2(t_0 - s_0)}{\varepsilon^2} + G\Big(\frac{2}{\varepsilon^2}(x_0 - y_0) - 2\varepsilon y_0,\ y_0\Big) \geq 0. \tag{2.38}$$

Step 6: Contradiction. Subtract (2.37) from (2.38) and drop the non-negative terms $\eta/(T-t_0)^2$, $\eta/(T-s_0)^2$ on the left:

$$2\lambda \leq 2\lambda + \frac{\eta}{(T-t_0)^2} + \frac{\eta}{(T-s_0)^2} \leq G(p_2, y_0) - G(p_1, x_0),$$

where $p_1 := \frac{2}{\varepsilon^2}(x_0 - y_0) + 2\varepsilon x_0$ and $p_2 := \frac{2}{\varepsilon^2}(x_0 - y_0) - 2\varepsilon y_0$. Then $|p_2 - p_1| = 2\varepsilon|x_0 + y_0| \leq 2\varepsilon(|x_0| + |y_0|)$ and $|p_1| \leq \frac{2}{\varepsilon^2}|x_0 - y_0| + 2\varepsilon|x_0|$. By (2.3) and (2.4),

$$\begin{aligned} G(p_2, y_0) - G(p_1, x_0) &\leq |G(p_2, y_0) - G(p_1, y_0)| + |G(p_1, y_0) - G(p_1, x_0)| \\ &\leq C|p_2 - p_1| + C|x_0 - y_0|(1 + |p_1|). \end{aligned}$$

Combining the last two displays (and enlarging C),

$$\lambda \leq C\varepsilon(|x_0| + |y_0|) + C|x_0 - y_0|\Big(1 + \frac{|x_0 - y_0|}{\varepsilon^2} + \varepsilon(|x_0| + |y_0|)\Big). \tag{2.39}$$

Now let $\varepsilon \to 0$. By (2.33), $\varepsilon(|x_0| + |y_0|) \to 0$. By (2.35), $|x_0 - y_0| \to 0$ and $|x_0 - y_0|^2/\varepsilon^2 = o(\varepsilon)^2/\varepsilon^2 \to 0$, and $|x_0 - y_0|\,\varepsilon(|x_0| + |y_0|) \to 0$. Hence the right-hand side of (2.39) tends to 0, and we obtain $0 < \lambda \leq 0$. This contradiction proves $u \leq \tilde{u}$, and by symmetry $u = \tilde{u}$. □

Corollary 2.18. *Let H be as in (2.1) and let g be bounded and Lipschitz. Then the terminal-value problem (2.7) has at most one viscosity solution in the sense of Definition 2.8.*

Proof. By Proposition 2.5 and Lemma 2.9, $-H$ is a Hamiltonian in the sense of Definition 2.3. Let u_1, u_2 be viscosity solutions of (2.7). By Proposition 2.10, $w_i(x,t) := u_i(x, T-t)$ are viscosity solutions of (2.8), that is, of (2.6) with $G = -H$ and the same initial data g. Theorem 2.17 gives $w_1 = w_2$, hence $u_1 = u_2$. □

2.6 The Eikonal equation (revisited)

We return to Example 2.6 and show that, among its generalized solutions, viscosity solutions distinguish the distance function.

The Eikonal equation as a Hamilton-Jacobi equation

The Eikonal equation (2.5) is $G(u') = 0$ with $G(p) = |p| - 1$. This G is a Hamiltonian in the sense of Definition 2.3: $|G(p) - G(q)| \leq |p - q|$, and since G does not depend on x, (2.4) holds trivially. The equation differs from the setting of Definition 2.7 in two ways: it is stationary (there is no time variable), and it is posed on a bounded interval with boundary values. The definition adapts by dropping t: a continuous function $u : [0,1] \to \mathbb{R}$ is a *viscosity solution* of (2.5) if $u(0) = u(1) = 0$ and, for every $v \in C^\infty(0,1)$,

- if $u - v$ has a local maximum at $x_0 \in (0,1)$, then $|v'(x_0)| - 1 \leq 0$;

- if $u - v$ has a local minimum at $x_0 \in (0,1)$, then $|v'(x_0)| - 1 \geq 0$.

Since the arguments below use only that v is differentiable at x_0, they hold equally for C^1 test functions. Theorem 2.17 concerns the initial-value problem and is not used here; we only compare two candidate solutions directly.

The distance function is a viscosity solution

Proposition 2.19. *The function $u_0(x) = \min\{x, 1-x\}$ is a viscosity solution of (2.5).*

Proof. Clearly u_0 is continuous and $u_0(0) = u_0(1) = 0$. Let $v \in C^\infty(0,1)$.

Points $x_0 \neq \frac{1}{2}$. Here u_0 is differentiable with $u_0'(x_0) = \pm 1$. If $u_0 - v$ has a local maximum or minimum at x_0, then $(u_0 - v)'(x_0) = 0$, so $v'(x_0) = u_0'(x_0)$ and $|v'(x_0)| - 1 = 0$. Both conditions hold, exactly as in Proposition 2.12.

Local maximum at $x_0 = \frac{1}{2}$. Suppose $u_0 - v$ has a local maximum at $\frac{1}{2}$. There is $\delta > 0$ with

$$u_0(x) - v(x) \leq u_0(\tfrac{1}{2}) - v(\tfrac{1}{2}) \quad \Longleftrightarrow \quad v(x) - v(\tfrac{1}{2}) \geq u_0(x) - u_0(\tfrac{1}{2}) \qquad (|x - \tfrac{1}{2}| < \delta).$$

For $x \in (\frac{1}{2} - \delta, \frac{1}{2})$ we have $u_0(x) = x$ and $u_0(\frac{1}{2}) = \frac{1}{2}$, so $v(x) - v(\frac{1}{2}) \geq x - \frac{1}{2}$. Dividing by $x - \frac{1}{2} < 0$ reverses the inequality:

$$\frac{v(x) - v(\frac{1}{2})}{x - \frac{1}{2}} \leq 1.$$

Letting $x \to \frac{1}{2}^-$ gives $v'(\frac{1}{2}) \leq 1$. For $x \in (\frac{1}{2}, \frac{1}{2} + \delta)$ we have $u_0(x) = 1 - x$, so $v(x) - v(\frac{1}{2}) \geq -(x - \frac{1}{2})$. Dividing by $x - \frac{1}{2} > 0$,

$$\frac{v(x) - v(\frac{1}{2})}{x - \frac{1}{2}} \geq -1,$$

and letting $x \to \frac{1}{2}^+$ gives $v'(\frac{1}{2}) \geq -1$. Hence $|v'(\frac{1}{2})| \leq 1$, i.e. $|v'(\frac{1}{2})| - 1 \leq 0$.

Local minimum at $x_0 = \frac{1}{2}$. We show that this case cannot occur. Suppose $u_0 - v$ has a local minimum at $\frac{1}{2}$, so $v(x) - v(\frac{1}{2}) \leq u_0(x) - u_0(\frac{1}{2})$ near $\frac{1}{2}$. For $x < \frac{1}{2}$ this reads $v(x) - v(\frac{1}{2}) \leq x - \frac{1}{2}$, and dividing by $x - \frac{1}{2} < 0$ gives $\frac{v(x) - v(1/2)}{x - 1/2} \geq 1$, hence $v'(\frac{1}{2}) \geq 1$. For $x > \frac{1}{2}$ it reads $v(x) - v(\frac{1}{2}) \leq -(x - \frac{1}{2})$, hence $v'(\frac{1}{2}) \leq -1$. These are incompatible, so no such v exists and the super-solution condition holds vacuously. □

A generalized solution which is not a viscosity solution

Proposition 2.20. *Let $\mathcal{A} = \{0, \frac{1}{2}, 1\}$ and $u^{(2)}(x) = \operatorname{dist}(x, \mathcal{A}) = \min\{|x|, |x - \frac{1}{2}|, |x - 1|\}$. Then $u^{(2)}$ is a generalized solution of (2.5), but it is not a viscosity solution.*

Proof. Away from the points of $\mathcal{A}$ and the midpoints $\frac{1}{4}, \frac{3}{4}$, $u^{(2)}$ is differentiable with $|u^{(2)\prime}| = 1$, and $u^{(2)}(0) = u^{(2)}(1) = 0$. So it is a generalized solution.

For $x \in (\frac{1}{4}, \frac{3}{4})$ the nearest point of $\mathcal{A}$ is $\frac{1}{2}$, so $u^{(2)}(x) = |x - \frac{1}{2}|$ there. Take $v \equiv 0$, which belongs to $C^\infty(0,1)$ with $v' \equiv 0$. For $x \in (\frac{1}{4}, \frac{3}{4})$,

$$(u^{(2)} - v)(x) = |x - \tfrac{1}{2}| \geq 0 = (u^{(2)} - v)(\tfrac{1}{2}),$$

so $u^{(2)} - v$ has a local minimum at $\frac{1}{2}$. The super-solution condition would require $|v'(\frac{1}{2})| - 1 \geq 0$, that is $0 - 1 \geq 0$, which is false. Hence $u^{(2)}$ is not a viscosity solution. □

Remark 2.21. The two cases differ in the type of kink. The distance function u_0 has a concave kink (a peak) at $\frac{1}{2}$: it can be touched from above by many smooth functions, all with slope in $[-1, 1]$, but from below by none. The function $u^{(2)}$ has a convex kink (a valley) at $\frac{1}{2}$ and is touched from below by $v \equiv 0$, whose slope violates the super-solution inequality. For the same reason $-u_0$, which is also a generalized solution, is not a viscosity solution. Viscosity solutions thus select the distance function out of the infinitely many generalized solutions.

Chapter 3

Viscosity Solutions and Value Function

3.1 The main theorem

Theorem 3.1 (PDE for the value function)**.** *The value function u defined in (1.5) is the unique viscosity solution of the terminal-value problem (2.7) for the Hamilton–Jacobi–Bellman equation.*

Proof. **Step 0: What has to be shown.** By Theorem 1.6, u is bounded and Lipschitz, hence uniformly continuous. It remains to verify the conditions (a), (b) of Definition 2.8. Uniquensee follows from Theorem 2.17

Step 1: Terminal condition. For $t = T$ the integral in (1.3) is over an empty interval, and $\mathbf{x}(T) = x$ regardless of the control. Thus $C_{x,T}[\alpha] = g(x)$ for all $\alpha \in \mathcal{A}$, and so

$$u(x,T) = \inf_{\alpha\in\mathcal{A}} C_{x,T}[\alpha] = g(x) \qquad (x \in \mathbb{R}^n).$$

Step 2: Sub-solution at a local maximum. Let $v \in C^\infty(\mathbb{R}^n \times (0,T))$ and suppose $u - v$ has a local maximum at $(x_0, t_0) \in \mathbb{R}^n \times (0,T)$. We must show

$$v_t(x_0,t_0) + \min_{a\in A}\Big\{f(x_0,a)\cdot Dv(x_0,t_0) + r(x_0,a)\Big\} \geq 0. \tag{3.1}$$

Suppose, to the contrary, that the left-hand side equals some $\Phi < 0$.

(i) A good constant control. Since A is compact and the expression in braces is continuous in a, the minimum is attained at some $a^* \in A$, so that

$$v_t(x_0,t_0) + f(x_0,a^*)\cdot Dv(x_0,t_0) + r(x_0,a^*) = \Phi < 0.$$

Set $\theta := -\Phi/2 > 0$. The map $(x,t) \mapsto v_t(x,t) + f(x,a^*)\cdot Dv(x,t) + r(x,a^*)$ is continuous (as v is smooth and f, r are continuous), so there is $\delta > 0$ with

$$v_t(x,t) + f(x,a^*)\cdot Dv(x,t) + r(x,a^*) \leq -\theta < 0 \quad \text{whenever } (x,t) \in N_\delta, \tag{3.2}$$

where $N_\delta := \{(x,t) : |x - x_0| + |t - t_0| < \delta\}$. Because $u - v$ has a local maximum at (x_0,t_0), we may shrink δ further so that also $N_\delta \subset \mathbb{R}^n \times (0,T)$ (possible since $0 < t_0 < T$) and

$$(u-v)(x,t) \leq (u-v)(x_0,t_0) \quad \text{for all } (x,t) \in N_\delta. \tag{3.3}$$

(ii) The trajectory stays in N_δ. Let $\alpha(s) \equiv a^*$ for $t_0 \le s \le T$ and let $\mathbf{x}(\cdot)$ solve

$$\dot{\mathbf{x}}(s) = f(\mathbf{x}(s), a^*) \ (t_0 < s < T), \qquad \mathbf{x}(t_0) = x_0.$$

Since $|f| \le M$, we have $|\mathbf{x}(s) - x_0| \le M(s - t_0)$. Fix $0 < h < \delta/(M+1)$. Then for $t_0 \le s \le t_0 + h$, $|\mathbf{x}(s) - x_0| + |s - t_0| \le (M+1)h < \delta$, so $(\mathbf{x}(s), s) \in N_\delta$. Hence (3.2) applies along the trajectory:

$$v_t(\mathbf{x}(s), s) + f(\mathbf{x}(s), a^*) \cdot Dv(\mathbf{x}(s), s) + r(\mathbf{x}(s), a^*) \le -\theta \qquad (t_0 \le s \le t_0 + h). \tag{3.4}$$

(iii) Upper bound from v. Since $(\mathbf{x}(t_0 + h), t_0 + h) \in N_\delta$, (3.3) gives $u(\mathbf{x}(t_0 + h), t_0 + h) - v(\mathbf{x}(t_0 + h), t_0 + h) \le u(x_0, t_0) - v(x_0, t_0)$, that is,

$$\begin{aligned} u(\mathbf{x}(t_0 + h), t_0 + h) - u(x_0, t_0) &\le v(\mathbf{x}(t_0 + h), t_0 + h) - v(x_0, t_0) \\ &= \int_{t_0}^{t_0+h} \frac{d}{ds} v(\mathbf{x}(s), s)\, ds \\ &= \int_{t_0}^{t_0+h} \Big[v_t(\mathbf{x}(s), s) + Dv(\mathbf{x}(s), s) \cdot \dot{\mathbf{x}}(s) \Big] ds \\ &= \int_{t_0}^{t_0+h} \Big[v_t(\mathbf{x}(s), s) + f(\mathbf{x}(s), a^*) \cdot Dv(\mathbf{x}(s), s) \Big] ds. \end{aligned} \tag{3.5}$$

Here we used the chain rule (valid since v is smooth and $\mathbf{x} \in C^1$ for a constant control) and the ODE $\dot{\mathbf{x}} = f(\mathbf{x}, a^*)$.

(iv) Lower bound from dynamic programming. The constant control a^* is one admissible control in (1.10) (with $(x, t) = (x_0, t_0)$), so the infimum is at most its value:

$$u(x_0, t_0) \le \int_{t_0}^{t_0+h} r(\mathbf{x}(s), a^*)\, ds + u(\mathbf{x}(t_0 + h), t_0 + h). \tag{3.6}$$

(v) Contradiction. Rearranging (3.6) and using (3.5) and then (3.4),

$$\begin{aligned} 0 &\le \int_{t_0}^{t_0+h} r(\mathbf{x}(s), a^*)\, ds + u(\mathbf{x}(t_0 + h), t_0 + h) - u(x_0, t_0) \\ &\le \int_{t_0}^{t_0+h} \Big[v_t(\mathbf{x}(s), s) + f(\mathbf{x}(s), a^*) \cdot Dv(\mathbf{x}(s), s) + r(\mathbf{x}(s), a^*) \Big] ds \ \le \ -\theta h < 0, \end{aligned}$$

which is absurd. Hence (3.1) holds.

Step 3: Super-solution at a local minimum. Let $v \in C^\infty(\mathbb{R}^n \times (0, T))$ and suppose $u - v$ has a local minimum at $(x_0, t_0) \in \mathbb{R}^n \times (0, T)$. We must show

$$v_t(x_0, t_0) + \min_{a \in A} \Big\{ f(x_0, a) \cdot Dv(x_0, t_0) + r(x_0, a) \Big\} \le 0. \tag{3.7}$$

Suppose not, so the left-hand side is strictly positive.

(i) A uniform positive lower bound. Define $\varphi(x, t, a) := v_t(x, t) + f(x, a) \cdot Dv(x, t) + r(x, a)$. By assumption $\min_{a \in A} \varphi(x_0, t_0, a) > 0$; choose $\theta > 0$ with $\varphi(x_0, t_0, a) \ge 2\theta$ for all $a \in A$. The function φ is continuous, hence uniformly continuous on the compact set $\overline{B} \times A$ for any compact neighbourhood $\overline{B}$ of (x_0, t_0). Therefore there is $\delta > 0$ such that

$$v_t(x, t) + f(x, a) \cdot Dv(x, t) + r(x, a) \ge \theta > 0 \quad \text{for all } a \in A, \ (x, t) \in N_\delta, \tag{3.8}$$

where $N_\delta = \{|x - x_0| + |t - t_0| < \delta\}$. As before we shrink δ so that $N_\delta \subset \mathbb{R}^n \times (0, T)$ and

$$(u - v)(x, t) \ge (u - v)(x_0, t_0) \quad \text{for all } (x, t) \in N_\delta. \tag{3.9}$$

(ii) Choice of h. Fix $0 < h < \delta/(M+1)$. For *every* admissible control α, the response $\mathbf{x}(\cdot)$ of

$$\dot{\mathbf{x}}(s) = f(\mathbf{x}(s), \alpha(s)) \ (t_0 < s < T), \qquad \mathbf{x}(t_0) = x_0$$

satisfies $|\mathbf{x}(s) - x_0| \le M(s - t_0)$, so $(\mathbf{x}(s), s) \in N_\delta$ for $t_0 \le s \le t_0 + h$. Thus h is independent of α. This uniformity is essential, since the control chosen in (iv) depends on h.

(iii) Lower bound from v. By (3.9), for any control α,

$$\begin{aligned} u(\mathbf{x}(t_0+h), t_0+h) - u(x_0, t_0) &\ge v(\mathbf{x}(t_0+h), t_0+h) - v(x_0, t_0) \\ &= \int_{t_0}^{t_0+h} \Big[v_t(\mathbf{x}(s), s) + f(\mathbf{x}(s), \alpha(s)) \cdot Dv(\mathbf{x}(s), s) \Big] ds. \end{aligned} \tag{3.10}$$

For a merely measurable control, $\mathbf{x}$ is Lipschitz (since $|\dot{\mathbf{x}}| \le M$), hence absolutely continuous, so $s \mapsto v(\mathbf{x}(s), s)$ is absolutely continuous with derivative $v_t + Dv \cdot \dot{\mathbf{x}}$ for almost every s, and $\dot{\mathbf{x}}(s) = f(\mathbf{x}(s), \alpha(s))$ almost everywhere. The fundamental theorem of calculus therefore applies.

(iv) Upper bound from dynamic programming. By (1.10) and the definition of the infimum, there exists a control $\alpha \in \mathcal{A}$ such that the quantity in braces is within $\theta h/2$ of $u(x_0, t_0)$, i.e.

$$u(x_0, t_0) \ge \int_{t_0}^{t_0+h} r(\mathbf{x}(s), \alpha(s))\, ds + u(\mathbf{x}(t_0+h), t_0+h) - \frac{\theta h}{2}. \tag{3.11}$$

We use this particular α (and its trajectory) in (3.10).

(v) Contradiction. From (3.11), $\int_{t_0}^{t_0+h} r\, ds + u(\mathbf{x}(t_0+h), t_0+h) - u(x_0, t_0) \le \theta h/2$. Combining with (3.10) and (3.8) (valid for all $a \in A$, in particular for $a = \alpha(s)$),

$$\frac{\theta h}{2} \ \ge \ \int_{t_0}^{t_0+h} \Big[v_t(\mathbf{x}(s), s) + f(\mathbf{x}(s), \alpha(s)) \cdot Dv(\mathbf{x}(s), s) + r(\mathbf{x}(s), \alpha(s)) \Big] ds \ \ge \ \theta h,$$

which is impossible because $\theta h > \theta h/2$. Hence (3.7) holds.

Step 4: Uniqueness. By Steps 1–3, u is a viscosity solution of (2.7). By (1.4), g is bounded and Lipschitz, and H is a Hamiltonian by Proposition 2.5. Hence Corollary 2.18 shows that any viscosity solution of (2.7) coincides with u. □

3.2 Design of optimal controls (revisited)

In Section 1.4 we designed an optimal feedback control assuming $u \in C^1$. We keep the same construction: fix $0 < t \le T$ and $x \in \mathbb{R}^n$, and consider

$$\begin{cases} \dot{\mathbf{x}}^*(s) = f(\mathbf{x}^*(s), \alpha^*(s)) \quad (t < s < T), \\ \mathbf{x}^*(t) = x, \end{cases} \tag{3.12}$$

where $\alpha^*(s) \in A$ is chosen to attain the minimum in the Hamiltonian,

$$f(\mathbf{x}^*(s), \alpha^*(s)) \cdot Du(\mathbf{x}^*(s), s) + r(\mathbf{x}^*(s), \alpha^*(s)) = H\Big(Du(\mathbf{x}^*(s), s), \mathbf{x}^*(s)\Big). \tag{3.13}$$

What changes now is that u is only Lipschitz (Theorem 1.6), so two points of the earlier argument need care.

1. *The gradient.* By Rademacher's theorem, Du exists almost everywhere in $\mathbb{R}^n \times (0,T)$, and by Theorem 2.14 (transferred to the terminal-value problem by Proposition 2.10) the equation $u_t + H(Du, x) = 0$ holds at every point of differentiability. However, the curve $s \mapsto (\mathbf{x}^*(s), s)$ has measure zero in $\mathbb{R}^n \times (0,T)$, so "almost everywhere" gives no information along it. Thus (3.13) makes sense only if u is differentiable at the points of the trajectory, and this must be assumed.

2. *The chain rule.* For $u \in C^1$ the derivative of $s \mapsto u(\mathbf{x}^*(s), s)$ was computed by the classical chain rule. Now $s \mapsto u(\mathbf{x}^*(s), s)$ is only Lipschitz, being a composition of the Lipschitz function u with the Lipschitz curve $\mathbf{x}^*$ (recall $|\dot{\mathbf{x}}^*| \le M$). It is therefore absolutely continuous, and the fundamental theorem of calculus still applies, but the chain rule can be used only at those s where u is differentiable at $(\mathbf{x}^*(s), s)$.

With these two points taken care of, the verification of Section 1.4 goes through.

Proposition 3.2 (Optimality of the feedback control)**.** *Let $\alpha^*(\cdot)$ be a measurable control with response $\mathbf{x}^*(\cdot)$ as in (3.12), and suppose that for almost every $s \in (t,T)$ the function u is differentiable at $(\mathbf{x}^*(s), s)$ and (3.13) holds. Then $u(x,t) = C_{x,t}[\alpha^*(\cdot)]$, so α^* is an optimal control.*

Proof. Let $s \in (t,T)$ be a point as in the hypothesis at which $\mathbf{x}^*$ is also differentiable; this holds for almost every s. By the consistency theorem, $u_t + H(Du, x) = 0$ at $(\mathbf{x}^*(s), s)$, and by the chain rule at a point of differentiability of u, together with (3.12) and (3.13),

$$\begin{aligned} \frac{d}{ds} u(\mathbf{x}^*(s), s) &= u_t(\mathbf{x}^*(s), s) + Du(\mathbf{x}^*(s), s) \cdot f(\mathbf{x}^*(s), \alpha^*(s)) \\ &= u_t(\mathbf{x}^*(s), s) + H\Big(Du(\mathbf{x}^*(s), s), \mathbf{x}^*(s)\Big) - r(\mathbf{x}^*(s), \alpha^*(s)) \\ &= -\, r(\mathbf{x}^*(s), \alpha^*(s)). \end{aligned}$$

Since $s \mapsto u(\mathbf{x}^*(s), s)$ is absolutely continuous, we may integrate from t to T and use $u(\cdot, T) = g$:

$$g(\mathbf{x}^*(T)) - u(x,t) = -\int_t^T r(\mathbf{x}^*(s), \alpha^*(s))\, ds,$$

that is, $u(x,t) = C_{x,t}[\alpha^*(\cdot)]$. Hence α^* attains the infimum in (1.5). □

Remark 3.3. The hypothesis of Proposition 3.2 is not automatic: a Lipschitz u may fail to be differentiable along the trajectory, and a measurable selection α^* in (3.13) must also be available. Viscosity solutions guarantee that the value function is well defined and unique, but constructing optimal controls from it in general requires additional information on the structure of the non-differentiability set of u, which we do not pursue here.

Conclusion

In this report we studied how the value function of an optimal control problem is characterized by a Hamilton-Jacobi-Bellman equation.

In Chapter 1 we showed that the value function is bounded and Lipschitz continuous (Theorem 1.6) and satisfies the Dynamic Programming Principle (Theorem 1.7). Assuming that it is C^1, the Dynamic Programming Principle gives the HJB equation (Theorem 1.8), and a solution of this equation produces an optimal feedback control (Theorem 1.9).

The C^1 assumption is not satisfied in general (Example 2.1), and almost everywhere solutions do not determine the solution uniquely (Example 2.6). In Chapter 2 we therefore introduced viscosity solutions. We showed that they agree with classical solutions when these exist (Proposition 2.12), that a viscosity solution satisfies the equation at every point of differentiability (Theorem 2.14), and that terminal-value problems reduce to initial-value problems by time reversal (Proposition 2.10). The key result is the uniqueness of viscosity solutions (Theorem 2.17).

In Chapter 3 these ingredients combine into the main result, Theorem 3.1: the value function is the unique viscosity solution of the terminal-value problem for the HJB equation. The two halves of the argument are complementary. The Dynamic Programming Principle shows that the value function is a viscosity solution, and the uniqueness theorem shows that it is the only one. Thus the HJB equation determines the value function even when it is not differentiable.

The construction of optimal controls is less complete. When u is merely Lipschitz, Rademacher's theorem gives the gradient only almost everywhere in space-time, which says nothing along a single trajectory. The feedback construction therefore works only under the additional assumption that u is differentiable along the trajectory (Proposition 3.2).

Several natural directions remain open. One is to find conditions, such as semiconcavity of the value function, under which this differentiability holds, and hence to obtain optimal controls from viscosity solutions in general. Another is the case of Hamiltonians that do not depend on x and are convex, where explicit representation formulas such as the Hopf-Lax formula are available. A third is to prove the comparison principle for viscosity solutions in more general settings, or to study numerical schemes whose limits are viscosity solutions.

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